technology to develop electronic synapses that work just like the human brain, to power neural networks of the future, where computers are capable of understanding and processing the information as well, and as quickly as a human being can.

The brain's neurons encode information in the patterns and timing of spikes of activity. That encoding is hard to model using electronic hardware because most electronics use binary (0 and 1) switches. However, researchers have combined memristors and capacitors in a way that allows for the creation of spiking output patterns.

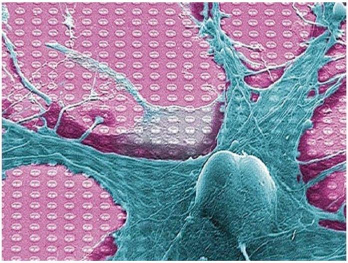
Memristors are devices made of materials that behave as insulators until these are heated, at which point these act as conductors. Researchers paired a memristor and a capacitor in a parallel circuit and applied a current. As the voltage heated it, the memristor behaved as a resistor until it reached a critical temperature; then it became a conductor. That switching allowed for full release of the energy stored in the capacitor and, thus, mimicked the spiking behaviour of neurons.

The system, which the researchers termed a neuristor, is a very simplified model of neuron behaviour and produces a much more regular spiking pattern than a real neuron. They believe that using a different memristor and a more complicated circuit could allow them to more closely reproduce neuron behaviour on a computer chip.

Neuristors

A neuristor is the simplest possible device that can capture the essential property of a neuron, that is, the ability to generate a spike or impulse of activity when a threshold is exceeded. A neuristor can be thought of as a slightly-leaky balloon that receives inputs in the form of puffs of air. The only major difference is that more complex neuristors can repeat the process again and again, as long as spikes occur no faster than a certain recharge period known as the refractory period.

A neuristor uses a relatively simple electronic circuit to generate spikes. Incoming signals charge a capacitor that is placed in parallel with a memristor. When that happens, charge builds up on the capacitor by incoming spikes' discharges, and we have a spiking neuron that comprises just two elementary circuit elements.



Neuristors are brain-like computer chips (Image courtesy: www.neoteo.com)

Details of trying to build neuristors have been largely unglamorous. In fact, most people trying to create artificial brains with neuristor-like elements have been content to do so by simulating these on regular old digital computers. A group has come up with an improved method of fabricating a neuristor made from niobium-dioxide (NbO_2). For now, their prototypes are too inefficient to be used in large numbers on a single chip, but the group is developing ways to make these more compatible with current fabrication methods.

This new study does not deny long-standing efforts of the neuromorphic community to build analogue neurons into chips. It is simply the next step towards getting people on board with neuristor hardware instead of neuristor simulation.

In order to mimic the amazing human brain, we need a chip and its constituting elements behaving like the constituting elements of the brain—transistors for neurons. A transistor in some ways behaves like a synapse, acting as a signal gate. When two neurons are in connection, electrochemical reactions through neurotransmitters relay specific signals. In a real synapse, calcium ions induce chemical signaling. Transistors use oxygen ions instead, engulfed in an 80-nanometre-thick layer of samarium-nickelate crystal, which is the analogue to the synapse channel.

When a voltage is applied to the crystal, oxygen ions slip through, changing the conductive properties of the lattice and altering signal-relaying capabilities. Strength of the connection is based on the time delay in the electric signal fed into it. It is the same way for real neurons that get stronger as these relay more signals. Exploiting unusual properties in modern materials, the synaptic transistor could mark the beginning of a new kind of artificial intelligence—one embedded not in smart algorithms but in the very architecture of a computer.

Each time a neuron initiates an action and another neuron reacts, the synapse between these increases the strength of its connection; the faster the neurons spike each